# Generated report 2: pathmnist

## Objective (as given)

"Balance the rarest tissue class to improve a classifier"

## Agent's decision

Task: class_balancing | decision: proceed | task run: class_balancing

profile.json shows 89,996 labelled images (well above the 200-sample floor), labels for every image (has_labels true, 9 classes), and imbalance_ratio 1.634, which is above the 1.5 threshold but mild: class 6 has 7,886 images versus 12,885 in the largest class (8). Because the imbalance is mild and class 6 already has thousands of real examples, a simple baseline (random oversampling of real class-6 images) may already do as well as synthesis; the evaluation therefore compares synthetic balancing against real-only and resampling baselines, and the honest expected outcome is a small or null effect. Generative modelling is still a reasonable thing to test because class-conditional image generators can add stain and texture variation that duplication cannot.

## Models run (headline metrics; details on the metrics page)

Run names are <model id>_s<seed>: cvae_s0 is model cvae trained with seed 0. Models in this report: cvae (seed 0); cgan (seeds 0, 1); cdiff (seed 0). The next page explains seeds, the evaluation space, references, and memorization.

```
cvae_s0          vae       FD 264.6948 (ref 0.2915)  C2ST 0.9998 (ref 0.5081)  label agr. 0.1465
cgan_s0          gan       FD 58.0703 (ref 0.2915)  C2ST 0.999 (ref 0.5081)  label agr. 0.3797
cgan_s1          gan       FD 58.852 (ref 0.1627)  C2ST 0.9986 (ref 0.502)  label agr. 0.3182
cdiff_s0         diffusion FD 42.6555 (ref 0.2915)  C2ST 0.9941 (ref 0.5081)  label agr. 0.8027
```

## Objective test: classifier on held-out real images

```
Target class(es) 6; real-only seed spread (noise level) 0.0063
real_seed0                   target F1 0.9175  macro F1 0.9337
real_seed1                   target F1 0.9112  macro F1 0.9358
real_plus_resampled          target F1 0.9191  macro F1 0.9366
real_plus_cvae_s0            target F1 0.9028  macro F1 0.933
real_plus_cgan_s0            target F1 0.8971  macro F1 0.9309
real_plus_cgan_s1            target F1 0.9  macro F1 0.9312
real_plus_cdiff_s0           target F1 0.9003  macro F1 0.9291
```

## Downstream-use verdicts

- [not_recommended] Top up class 6 (normal mucosa) with diffusion images (cdiff_s0)
- [not_recommended] Top up class 6 with conditional-GAN images (cgan_s0)
- [not_recommended] Top up class 6 with conditional-VAE images (cvae_s0)
- [conditional] Oversample real class-6 images (non-generative baseline) (real_plus_resampled)

## Summary (agent-written)

# Generated report 2: pathmnist (cont.)

The objective was to balance the rarest tissue class in PathMNIST to improve a classifier. The rarest class is class 6 (normal colon mucosa), and the imbalance is mild (1.634x largest to smallest). The agent trained a conditional VAE, a conditional GAN (two seeds) and a conditional diffusion model on a 20,000-image stratified subsample, topped class 6 up with 889 synthetic images from each, and trained the same classifier on each training set. None of the synthetic top-ups helped. Class-6 F1 on held-out real images was 0.9003 to 0.9028 with synthetic images, against 0.9191 for simple oversampling of real class-6 images and 0.9175 / 0.9112 for real data alone (noise level 0.0063). Bottom line: don't use these generators for class balancing. Oversampling is at least as good, and at this mild imbalance it is barely distinguishable from doing nothing.

# How to read this report

## Run names and seeds

Run names are <model id>_s<seed>: cvae_s0 is model cvae trained with seed 0. Models in this report: cvae (seed 0); cgan (seeds 0, 1); cdiff (seed 0).

A seed fixes the model's random initialization, mini-batch order, the validation slice used for early stopping, and the random draws when sampling; the held-out split and preprocessing are the same for every run. Models with two seeds are trained twice as a stability check: a conclusion that holds for both seeds does not depend on one lucky or unlucky initialization. Two runs are far too few to estimate run-to-run variability, so the spread between seeds is not a confidence interval and should not be read as one.

## Where realism is measured: the evaluation space

Images are compared in the 128-dimensional feature space of a small CNN classifier trained on the real training images (held-out accuracy 0.94). This plays the role that Inception features play in the usual FID: realism is judged by features that matter for recognizing these images. An ImageNet network would be a poor judge of small, domain-specific images, and too heavy to run here. The classifier never sees synthetic or held-out images during training, and the same classifier scores every model.

Its features only capture what helps tell the classes apart, so two image sets can look alike to it and still differ in ways it ignores. The sample grid figure is the visual check. memorization_pixels repeats the DCR check on raw pixels.

## Reference values

Every realism metric is shown with a real-vs-real reference: the same metric computed between a subsample of the training data and the held-out real data. That is the value two genuinely real samples of this size give, so a model is judged by how close it gets to the reference, not to a theoretical ideal.

## How memorization is characterized

For each sample, DCR (distance to closest record) is the distance to its nearest training sample in the evaluation space. The held-out real samples give the yardstick: how close a new sample from the same population typically lands to some training sample. dcr_ratio compares the median synthetic DCR to the median held-out DCR (about 1 = as far as new real data; well below 1 = suspiciously close). too_close_fraction counts synthetic samples closer than the 5th percentile of held-out DCRs (about 0.05 is expected without copying).

This detects near-copies of training samples. It is not a privacy guarantee: it does not test membership inference, attribute inference, or any formal property such as differential privacy.

## Do generated images belong to the requested class?

label_agreement is the share of synthetic images that the evaluation classifier assigns to the class they were generated for. Its reference is the classifier's accuracy on held-out real images, which is what a perfect class-conditional generator would reach. Low agreement means the model's class conditioning is weak, which matters directly for class balancing.

# How to read this report (cont.)

## The class-balancing test

The same classifier is trained on several training sets and scored on held-out real images: real data only (twice, with two seeds), real data with the target class oversampled by duplication, and real data with the target class topped up with each model's synthetic images. Synthetic images never enter the held-out set. The difference between the two real-only runs is the noise level; an improvement smaller than that cannot be told apart from training noise. Two runs give a rough yardstick, not a confidence interval.

# Objective interpretation

## Interpretation

Primary task: class_balancing

Secondary tasks: none

'Balance' and 'rarest tissue class' map directly to class_balancing; 'to improve a classifier' names the success measure as downstream classifier performance on the target class. profile.json lists minority_class = 6 (7,886 images), which DATA_DESCRIPTION.md names as normal colon mucosa, so target.class is set explicitly to '6' rather than 'auto' to make the choice auditable.

## Success criterion

On held-out real PathMNIST images, a classifier trained on real training data plus synthetic class-6 (normal colon mucosa) images achieves higher class-6 F1 than the same classifier trained on real data plus simple random oversampling of class 6, without lowering macro-F1.

## Whether to model, and how

Decision: proceed (task actually run: class_balancing)

profile.json shows 89,996 labelled images (well above the 200-sample floor), labels for every image (has_labels true, 9 classes), and imbalance_ratio 1.634, which is above the 1.5 threshold but mild: class 6 has 7,886 images versus 12,885 in the largest class (8). Because the imbalance is mild and class 6 already has thousands of real examples, a simple baseline (random oversampling of real class-6 images) may already do as well as synthesis; the evaluation therefore compares synthetic balancing against real-only and resampling baselines, and the honest expected outcome is a small or null effect. Generative modelling is still a reasonable thing to test because class-conditional image generators can add stain and texture variation that duplication cannot.

## Other readings considered

- 'Rarest' is nearly a tie: class 6 (normal colon mucosa, 7,886) and class 4 (mucus, 8,006) differ by 120 images (about 1.5%). I took the literal smallest class, 6, from profile.minority_class; balancing class 4 instead, or both, would be equally defensible and is not tested here.

- 'Balance' could mean topping class 6 up to the largest class (12,885) or to the median class size; the plan's synthetic count decides this and is a judgment call recorded there.

- 'Improve a classifier' does not name a classifier; the pipeline's fixed downstream classifier is used, and results may not transfer to a CNN trained at full scale.

- The objective could be read as general augmentation of all classes; it was not, because it names a single rarest class.

# Data and preprocessing

## Dataset (from DATA_DESCRIPTION.md)

PathMNIST, part of the MedMNIST v2 collection. Derived from a histopathology study of colorectal cancer: 28x28 RGB image patches extracted from hematoxylin and eosin (H&E) stained histology slides, downsampled from an original resolution of 224x224. Source: NCT-CRC-HE-100K (train/validation split) and CRC-VAL-HE-7K (test split, from a different clinical center; not loaded by this project). Only the train split is used.

## Profile

89,996 samples x 2,352 features; modality image (28, 28, 3)

Sparsity 0.0001; value range [0, 255]; non-negative integers: True

Labels: yes, 9 classes, imbalance ratio 1.634

## Held-out split

test_fraction 0.2, stratified: True -> 16000 training / 4000 held-out samples

Reason: After subsampling to 20,000 images, 0.2 holds out 4,000 real images (about 350 of the minority class 6, normal colon mucosa, at its 8.8% share in profile.json), enough to score class-6 F1 and run image two-sample tests, while leaving 16,000 for training. Stratified because labels exist for all 9 classes and class-6 F1 on held-out data is the success criterion, so the held-out set must keep the class mix.

## Preprocessing (applied identically for every model; fit on training data only)

- stratified_subsample {"n_samples": 20000}: profile.json has 89,996 images; at the measured ~12.5 s/epoch for conditional diffusion on ~14,000 images, the full set would allow only about 10 diffusion epochs inside the 20-minute budget shared with a CVAE and two GAN seeds. 20,000 stratified images (16,000 train, ~14,400 after the 10% validation hold-back) match the timing table and keep the 1.63x class mix, so class 6 stays the minority. Judgment call: more images per epoch versus more epochs; I chose more epochs because diffusion quality is epoch-limited at this budget.

- scale_unit_range : PathMNIST pixels are integers 0-255 (profile value_range [0, 255], mean 168); scaling to [0, 1] matches the Bernoulli CVAE decoder and the bounded generators, and keeps image metrics on an interpretable scale.

# Models, hyperparameters, and training

## cvae

Family: vae (conditional); input: transformed; seeds [0]; serves class_balancing; status planned; estimated 2.5 min/seed

Why this model: Fast, stable conditional baseline for generating class-6 (normal colon mucosa) patches; expected to give blurry means that lose gland texture, which is exactly what the sharper GAN and diffusion comparators test against. All three are judged by the same downstream class-6 F1 versus real-only and resampling baselines.

- latent_dim = 32: 32 dimensions for 28x28x3 H&E patches with 9 tissue classes and ~14,400 training images: enough to capture stain colour and coarse texture variation per class without the latent space fitting pixel noise.

- base_channels = 32: 32 conv channels keep the CVAE to ~2 s/epoch on ~14,400 images, so it serves as the cheap baseline within the 20-minute budget.

- hidden_dims = [256]: One 256-unit dense layer between the conv features and the 32-d latent suffices for 28x28 inputs where the conv stack already reduces the 2,352 pixels.

- likelihood = bernoulli: Bernoulli on [0,1]-scaled intensities: pixels are bounded after scale_unit_range, and a bounded decoder cannot produce out-of-range stain values.

- kl_warmup_epochs = 6: About 10% of the 60-epoch maximum, ramping the KL weight so the class-conditional decoder does not ignore the latent code early in training on these low-detail 28x28 patches.

- epochs = 60: 60 is a maximum with early stopping on validation loss; at ~2 s/epoch it costs about 2 minutes on 14,400 images.

- batch_size = 128: 128 gives ~112 steps per epoch on ~14,400 training images, enough gradient updates per epoch for stable early-stopping checks.

- lr = 0.001: Adam 1e-3 suits the small conv VAE; with ~6,700 total steps at most, a lower rate risks underfitting within 60 epochs.

Training (cvae_s0): 60 of 60 max epochs, kept epoch 57 (max epochs reached); 1.794 min on mps; 14400 fit / 1600 validation samples; 1,707,267 parameters

## cgan

Family: gan (conditional); input: transformed; seeds [0, 1]; serves class_balancing; status planned; estimated 3.5 min/seed

Why this model: Sharper samples than the CVAE, which may matter for class-6 mucosa where gland structure distinguishes it from adenocarcinoma epithelium (class 8). GANs are seed-sensitive, so two seeds show whether any downstream gain is reproducible rather than a lucky run.

- latent_dim = 64: 64-d noise for 9-class conditional generation of 28x28x3 tissue patches, giving room for within-class stain and texture variation.

- base_channels = 64: 64 channels give the generator enough capacity for H&E colour and

# Models, hyperparameters, and training (cont.)

glandular texture at 28x28; measured ~8 s/epoch on ~14,000 images.

- epochs = 25: 25 fixed epochs (GANs are not early-stopped) at ~8 s/epoch on ~14,400 images is ~3.3 minutes per seed, so two seeds fit alongside diffusion in the 20-minute budget.

- batch_size = 128: 128 gives ~112 discriminator/generator updates per epoch on ~14,400 images, ~2,800 per run.

- lr = 0.0002: 2e-4 with Adam is the DCGAN-stable rate; higher rates make the adversarial game on small-image data oscillate.

Training (cgan_s0): 25 of 25 max epochs, kept epoch 25 (planned epochs completed (GANs are not early-stopped)); 3.087 min on mps; 14400 fit / 1600 validation samples; 730,564 parameters

Training (cgan_s1): 25 of 25 max epochs, kept epoch 25 (planned epochs completed (GANs are not early-stopped)); 3.072 min on mps; 14400 fit / 1600 validation samples; 730,564 parameters

### cdiff

Family: diffusion (conditional); input: transformed; seeds [0]; serves class_balancing; status planned; estimated 9.5 min/seed

Why this model: Diffusion is the family most likely to produce realistic class-6 H&E texture and stain variation, which is the only way synthesis could beat plain duplication of the 7,886 real mucosa images. One seed because diffusion training is stable and the budget leaves no room for a second.

- timesteps = 400: 400 noise levels are sufficient for 28x28x3 images and keep the per-epoch cost at the measured ~12.5 s on ~14,000 images.

- base_channels = 32: 32-channel UNet is the size measured at ~12.5 s/epoch; larger would cut the epoch count the 20-minute budget allows.

- sampling_steps = 50: 50 DDIM steps take ~0.02 s per image, so the ~900 class-6 balancing images plus evaluation samples generate in well under a minute.

- guidance_scale = 3: 3.0, toward the upper end of the 2-4 range, because the goal is images that are reliably class 6 (normal mucosa) rather than drifting toward the similar adenocarcinoma class 8; judgment call, 2.0 would give more diversity at the cost of class fidelity.

- epochs = 40: 40 epochs at ~12.5 s/epoch on ~14,400 images is ~8.3 minutes, the largest share that fits the 20-minute budget after the CVAE (2.5) and two GAN seeds (7); early stopping applies.

- batch_size = 128: 128 gives ~112 denoising updates per epoch on ~14,400 images, ~4,500 over 40 epochs.

- lr = 0.0002: 2e-4 Adam is stable for a small UNet; diffusion is undertrained at this budget, and a higher rate risks instability more than it saves epochs.

# Models, hyperparameters, and training (cont.)

Training (cdiff_s0): 40 of 40 max epochs, kept epoch 40 (max epochs reached); 7.548 min on mps; 14400 fit / 1600 validation samples; 569,891 parameters

## Evaluation focus

The decision rests on metrics/downstream.json: class-6 F1 and macro-F1 on held-out real images for real-only (two seeds, giving the noise level), real + resampled class 6, and real + each model's synthetic class-6 images. Because the imbalance is mild (1.63x), the expected effect is small, and any gain must exceed the real-only seed spread and beat resampling. Image fidelity metrics (compared with their real-vs-real reference) and memorization distances explain why a model helps or not.

# Evaluation metrics (held-out real data)

Realism metrics use the 128-d feature space of the evaluation classifier (held-out accuracy 0.94); per-pixel statistics use all pixels. Each realism metric has a real-vs-real reference row.

## Fidelity

| | cvae_s0 | cgan_s0 | cgan_s1 | cdiff_s0 |
|---|---|---|---|---|
| n_compared | 4000 | 4000 | 4000 | 4000 |
| frechet_distance | 264.6948 | 58.0703 | 58.852 | 42.6555 |
| frechet_distance_reference | 0.2915 | 0.2915 | 0.1627 | 0.2915 |
| mmd2 | 0.37598 | 0.05428 | 0.05682 | 0.03936 |
| mmd2_reference | -0.0001 | -0.0001 | -0.00012 | -0.0001 |
| c2st_auc | 0.9998 | 0.999 | 0.9986 | 0.9941 |
| c2st_auc_reference | 0.5081 | 0.5081 | 0.502 | 0.5081 |
| precision | 0.674 | 0.7188 | 0.8815 | 0.6418 |
| recall | 0.034 | 0.0542 | 0.0583 | 0.288 |

## Conditional

| | cvae_s0 | cgan_s0 | cgan_s1 | cdiff_s0 |
|---|---|---|---|---|
| label_agreement | 0.1465 | 0.3797 | 0.3182 | 0.8027 |
| label_agreement_reference | 0.94 | 0.94 | 0.94 | 0.94 |

## Memorization

| | cvae_s0 | cgan_s0 | cgan_s1 | cdiff_s0 |
|---|---|---|---|---|
| dcr_median_synthetic | 5.0479 | 3.6608 | 3.8197 | 4.4458 |
| dcr_median_heldout | 2.4628 | 2.4628 | 2.4628 | 2.4628 |
| dcr_ratio | 2.0496 | 1.4864 | 1.5509 | 1.8052 |
| too_close_fraction | 0 | 0 | 0.0003 | 0 |

Evidence about copying of training samples, not a formal privacy guarantee.

## Memorization pixels

| | cvae_s0 | cgan_s0 | cgan_s1 | cdiff_s0 |
|---|---|---|---|---|
| dcr_median_synthetic | 1.4906 | 3.8141 | 4.0011 | 4.6275 |
| dcr_median_heldout | 4.1324 | 4.1324 | 4.1324 | 4.1324 |
| dcr_ratio | 0.3607 | 0.923 | 0.9682 | 1.1198 |
| too_close_fraction | 0.038 | 0 | 0.0008 | 0 |

DCR in raw pixel space (scaled to [0, 1]): detects near-identical copies of training images.

## Feature stats

| | cvae_s0 | cgan_s0 | cgan_s1 | cdiff_s0 |
|---|---|---|---|---|
| mean_corr | 0.9995 | 0.9719 | 0.965 | 0.9953 |
| var_corr | 0.9849 | 0.9718 | 0.977 | 0.9672 |

## Likelihood

| | cvae_s0 | cgan_s0 | cgan_s1 | cdiff_s0 |
|---|---|---|---|---|
| train_mean | -1397.421 | n/a | n/a | n/a |

# Evaluation metrics (held-out real data) (cont.)

| | | | | |
|---|---|---|---|---|
| heldout_mean | -1399.275 | n/a | n/a | n/a |
| per_dimension_heldout | -0.5949 | n/a | n/a | n/a |

Likelihoods are in each model's own input space and units; do not compare them across models with different input spaces.

# Objective test: class balancing on held-out real data

Target class(es): 6. Each was topped up to the largest training class (6: +889 images). Classifier: SmallCNN trained for exactly 20 epochs (cosine learning-rate decay, final weights) on each training set.

| training set | n_train | target F1 | macro F1 | accuracy |
|---|---|---|---|---|
| real_seed0 | 16000 | 0.9175 | 0.9337 | 0.9337 |
| real_seed1 | 16000 | 0.9112 | 0.9358 | 0.9355 |
| real_plus_resampled | 16889 | 0.9191 | 0.9366 | 0.937 |
| real_plus_cvae_s0 | 16889 | 0.9028 | 0.933 | 0.9335 |
| real_plus_cgan_s0 | 16889 | 0.8971 | 0.9309 | 0.9317 |
| real_plus_cgan_s1 | 16889 | 0.9 | 0.9312 | 0.932 |
| real_plus_cdiff_s0 | 16889 | 0.9003 | 0.9291 | 0.9295 |

Noise level (target-F1 difference between the two real-only seeds): 0.0063. noise_level_target_f1 is the target-class F1 difference between two real-only runs that differ only in the classifier's seed. A change smaller than this is indistinguishable from training noise. Two runs are a rough yardstick, not a confidence interval.

## Definitions

- target_f1: F1 score (harmonic mean of precision and recall) for the target class(es), on held-out real data.
- macro_f1: Unweighted mean of per-class F1 on held-out real data; checks that helping the target class did not hurt others.
- noise_level_target_f1: Target-class F1 difference between two real-only classifier runs with different seeds.

# How to read the metrics

- mmd2: Squared maximum mean discrepancy (RBF kernel, bandwidth = median distance among held-out real samples) between synthetic and held-out real samples in the evaluation PCA space. 0 = same distribution; compare with mmd2_reference. The estimator is unbiased, so values near 0 can be slightly negative.

- mmd2_reference: Same MMD, but a training-set subsample vs held-out real: the value two real samples of this size give.

- c2st_auc: Classifier two-sample test: cross-validated ROC AUC of a gradient-boosted classifier separating synthetic from held-out real samples. 0.5 = indistinguishable, 1.0 = perfectly separable.

- c2st_auc_reference: Same test, training subsample vs held-out real (should be near 0.5).

- precision: Fraction of synthetic samples inside the 5-NN radius of some held-out real sample (fidelity: are samples realistic?).

- recall: Fraction of held-out real samples inside the 5-NN radius of some synthetic sample (coverage: is the data's variety reproduced?).

- dcr_median_synthetic: DCR = distance to closest record: for one sample, the Euclidean distance (evaluation space) to its nearest training sample. This is the median DCR over synthetic samples.

- dcr_median_heldout: Median DCR over held-out real samples: how close a genuinely new sample from the same population typically is to some training sample. This is the yardstick for the synthetic DCR.

- dcr_ratio: dcr_median_synthetic / dcr_median_heldout. About 1 means synthetic samples are as far from the training data as new real samples are; well below 1 means they sit unusually close to training samples (possible copying); above 1 means they are farther away than real data (often a sign of off-distribution samples, not of privacy).

- too_close_fraction: Fraction of synthetic samples whose DCR is below the 5th percentile of held-out DCRs. If the model does not copy, about 0.05 is expected (the same share as for new real samples); much higher means a group of synthetic samples are near-copies of training samples. Evidence about copying only, not a privacy guarantee.

- mean_corr: Pearson correlation, across features, between per-feature means of synthetic and held-out real data (transformed space).

- var_corr: Same, for per-feature variances.

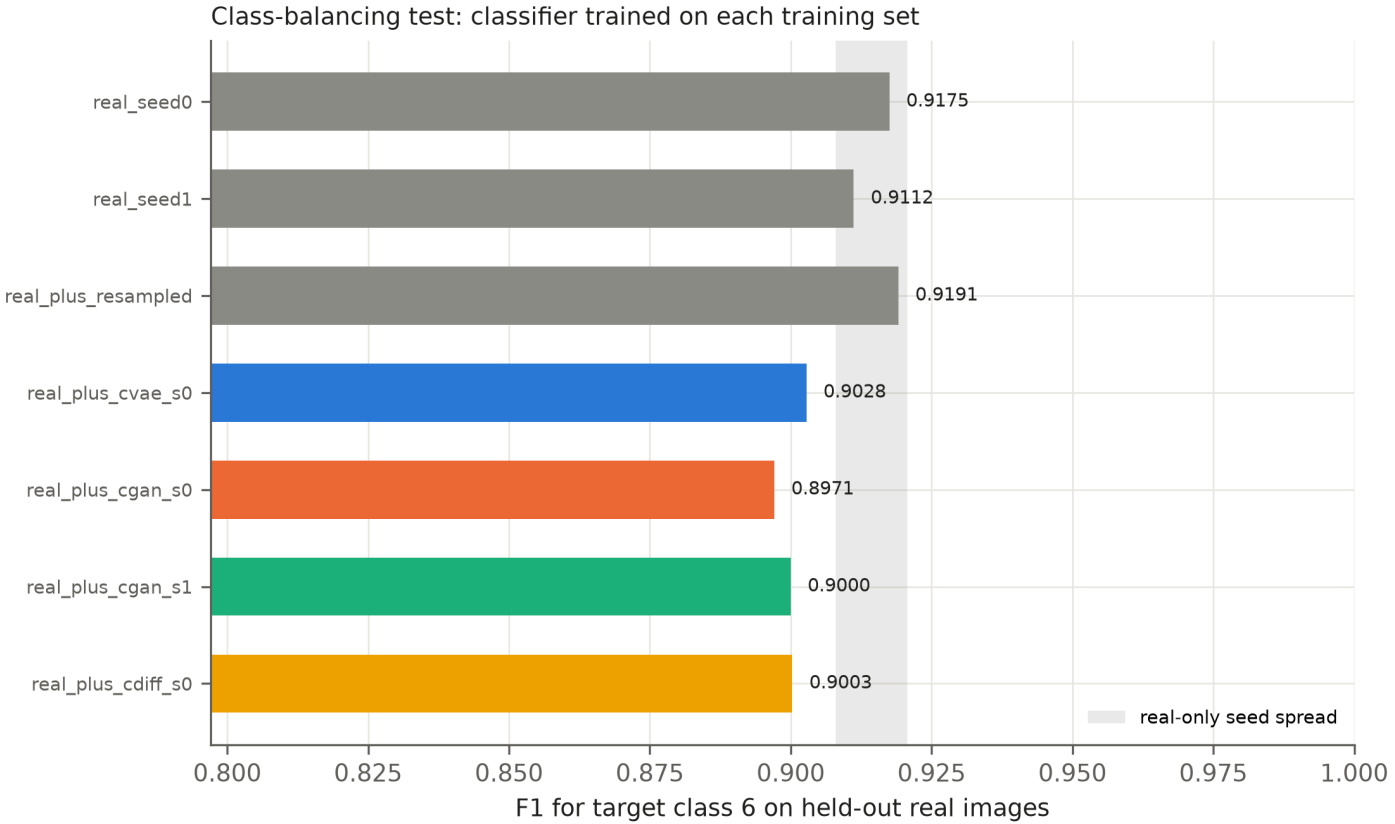
- train_mean: Mean per-sample ELBO (VAE, a lower bound) or exact log-likelihood (flow) on training data, in the model's own input space. Not comparable across models with different input spaces.

- heldout_mean: Same quantity on held-out real data; a large drop from train_mean indicates overfitting.

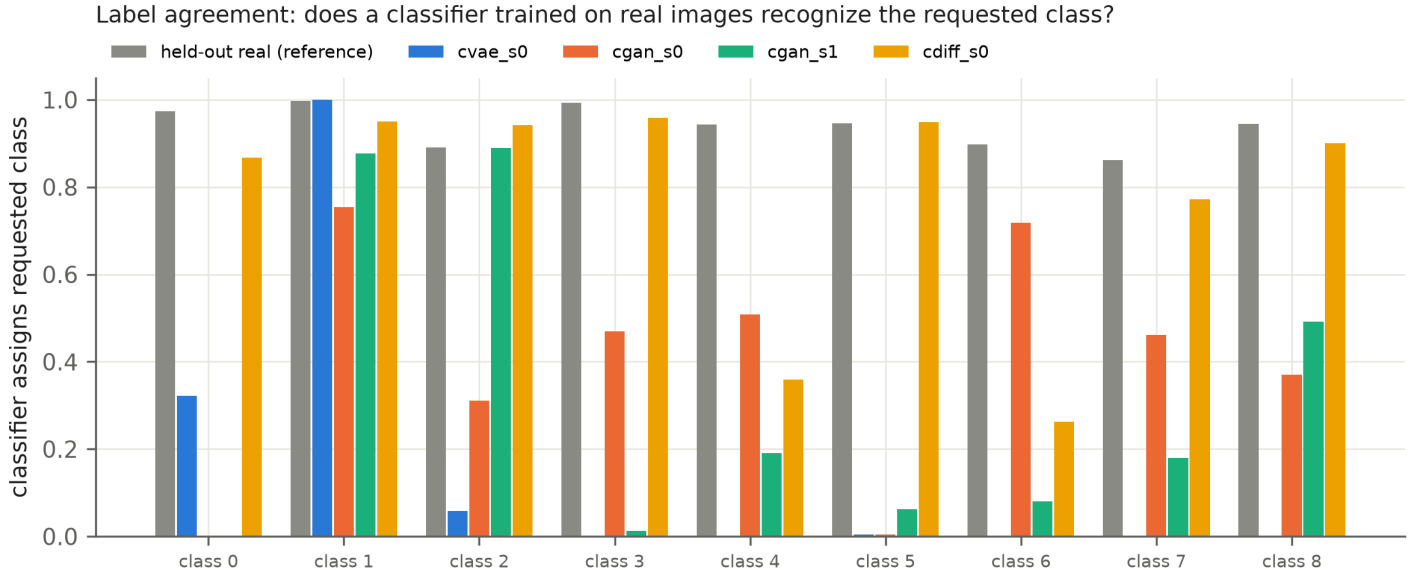
- frechet_distance: Frechet distance between Gaussians fit to synthetic and held-out real features (the FID formula, computed on the evaluation classifier's features instead of Inception's). 0 = same mean and covariance; compare with frechet_distance_reference.

# How to read the metrics (cont.)

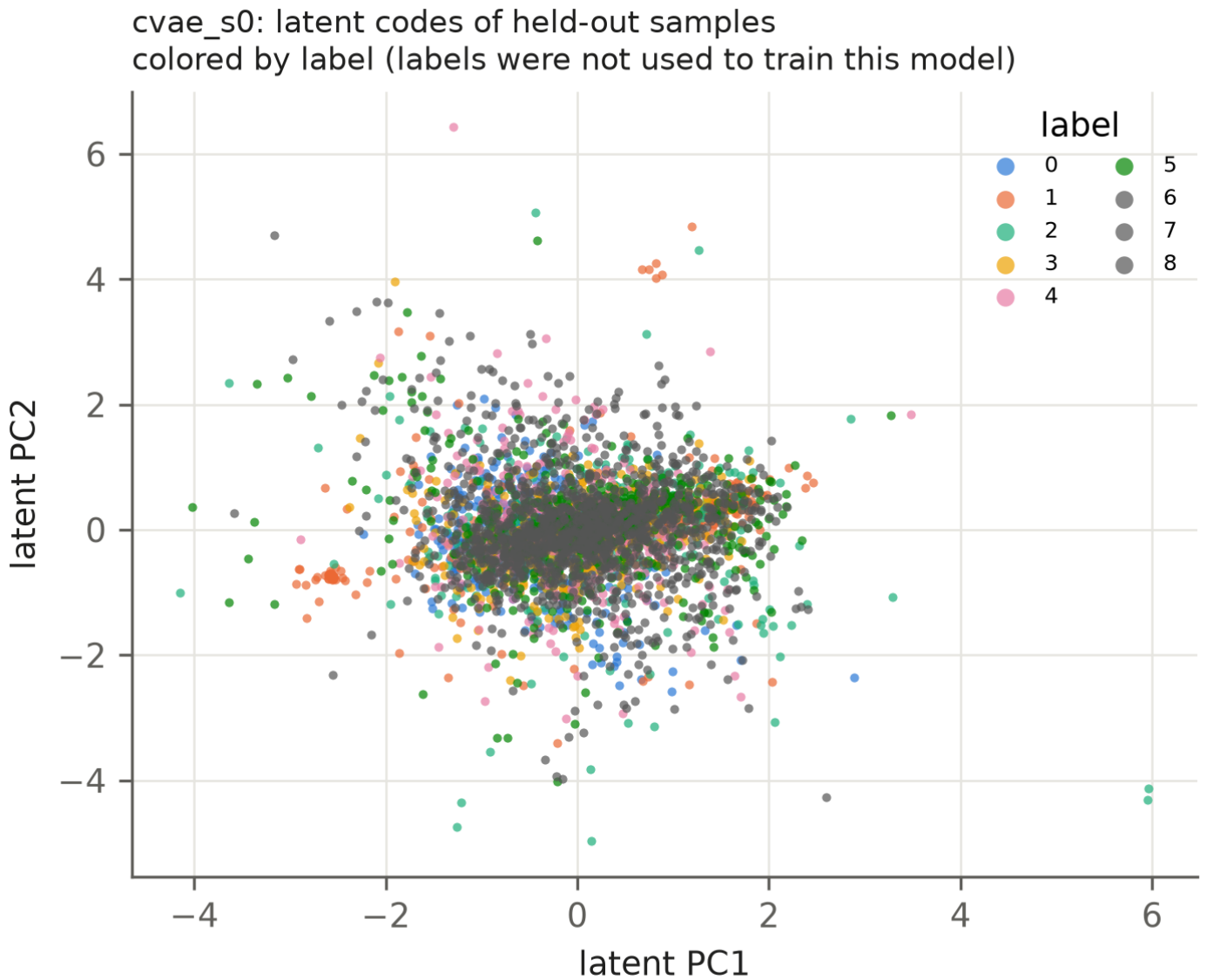
- frechet_distance_reference: Same distance, training-set subsample vs held-out real: what two real samples of this size give.
- label_agreement: Share of synthetic images that the evaluation classifier (trained on real images) assigns to the class they were generated for.
- label_agreement_reference: The classifier's accuracy on held-out real images: the agreement a perfect class-conditional generator would reach.




Class-balancing test: F1 for the target class on held-out real images, for a classifier trained on each training set.
The gray band spans the two real-only runs (training noise).

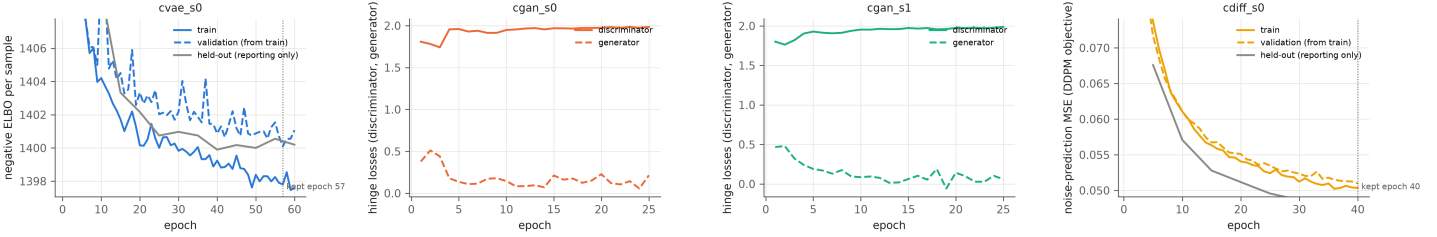



Per class: share of each model's images that the evaluation classifier assigns to the requested class; gray bars are the classifier's accuracy on held-out real images (the ceiling).

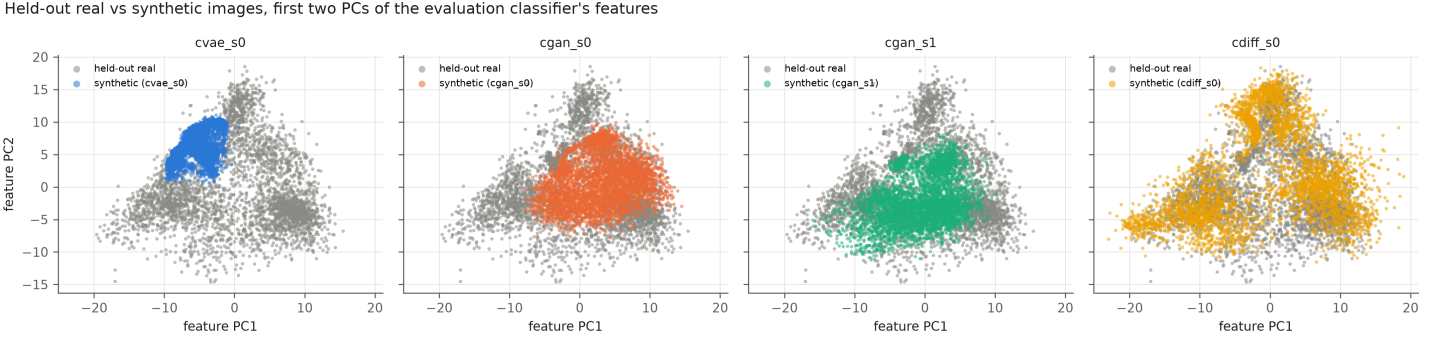



Two-dimensional PCA of a VAE's latent codes for held-out samples. A picture for orientation; neighbor-preservation numbers are on the metrics page.

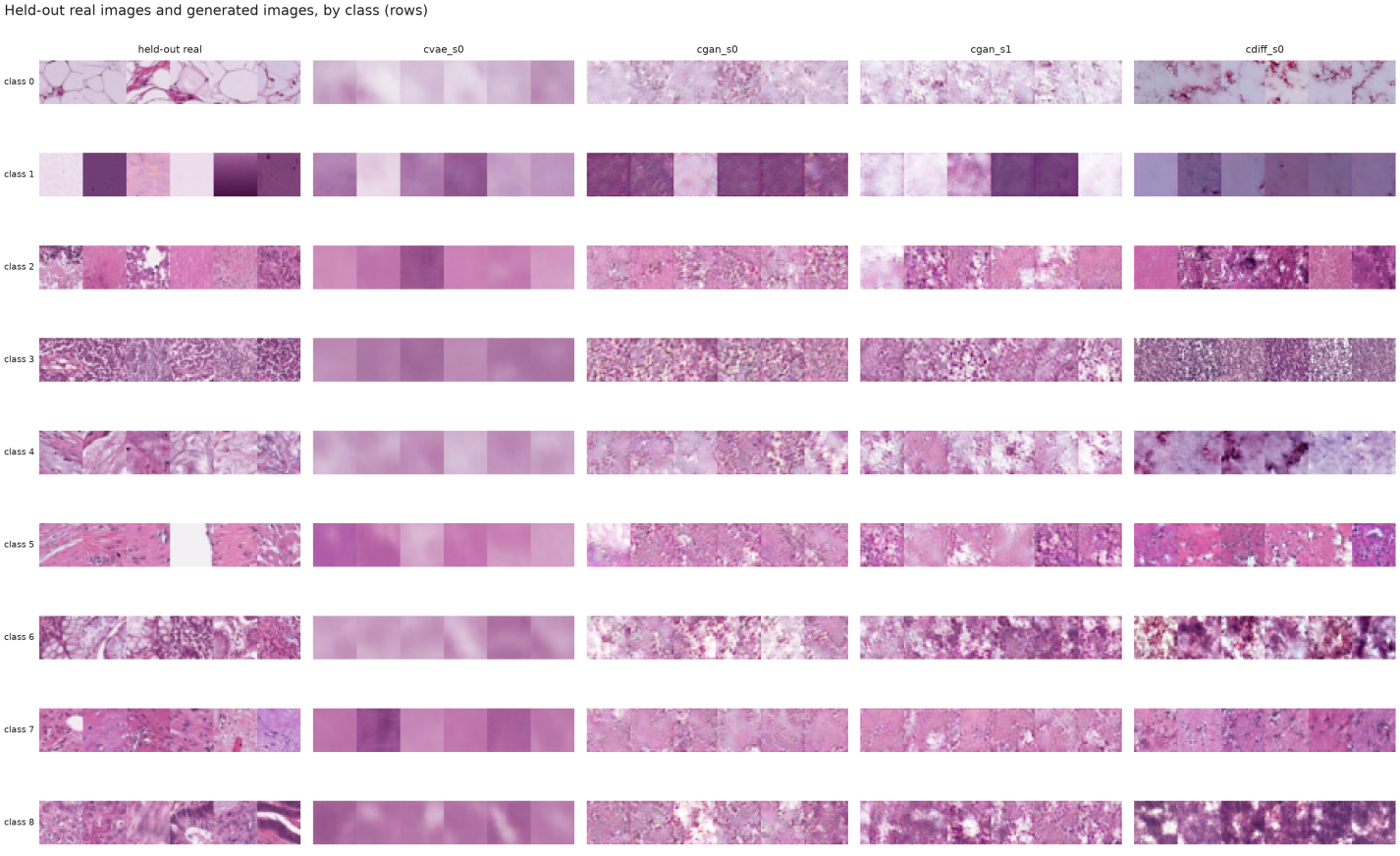
Training curves




Training curves. The validation set is 10% of the training split and decides early stopping; the held-out curve is shown for reference only and never used to choose anything.




Held-out real samples (gray) and synthetic samples in the first two principal components of the shared evaluation space (for images: of the evaluation classifier's features). Overlap in two components is necessary but not sufficient; the C2ST and MMD metrics use all components.




Held-out real images (first column block) and each model's samples, one row per class. Conditional models were asked for the class of the row; unconditional models' samples are not sorted by class.

# Findings

## What each model produced

All image fidelity numbers are computed in the 128-d feature space of a CNN trained on real training images (held-out accuracy 0.94). Each is given next to its real-vs-real reference.

**cvae_s0 (conditional VAE).** Frechet distance 264.6948 (reference 0.2915), C2ST AUC 0.9998 (reference 0.5081), precision 0.674, recall 0.034. Overall label agreement was 0.1465 (reference 0.94), and for class 6 it was 0.0: the classifier never recognised a generated class-6 image as mucosa. In pixel space the median distance to the nearest training image was low (dcr_ratio 0.3607, too_close_fraction 0.038). This reflects smooth, blurry images sitting near the average training image, not copies. The images show no texture (see figures). Training ran the full 60 epochs and kept epoch 57.

**cgan_s0 and cgan_s1 (conditional GAN, two seeds).** cgan_s0: Frechet distance 58.0703 (reference 0.2915), C2ST AUC 0.999 (reference 0.5081), precision 0.7188, recall 0.0542, label agreement 0.3797. cgan_s1: Frechet distance 58.852 (reference 0.1627), C2ST AUC 0.9986 (reference 0.502), precision 0.8815, recall 0.0583, label agreement 0.3182. For the target class, agreement was 0.7179 for cgan_s0 but 0.0798 for cgan_s1, so the seeds disagree sharply on whether they produce recognisable mucosa. Memorization: pixel dcr_ratio 0.923 and 0.9682, too_close_fraction 0.0 and 0.0008, so there are no signs of near-copies. Both seeds ran the fixed 25 epochs.

**cdiff_s0 (conditional diffusion).** This was the closest to real of the four: Frechet distance 42.6555 (reference 0.2915), C2ST AUC 0.9941 (reference 0.5081), precision 0.6418, recall 0.288, label agreement 0.8027 (reference 0.94). For class 6, however, agreement was only 0.2621, against held-out real class-6 accuracy of 0.8974. Class 4 (mucus) was also weak at 0.3596. Pixel dcr_ratio was 1.1198 with too_close_fraction 0.0. Training hit the 40-epoch maximum with the best epoch being the last one, so the model was still improving when the budget stopped it.

Even the best model was easy to tell apart from real images: every C2ST AUC is near 1 against references near 0.5.

## What the figures show

The sample grid shows the CVAE producing flat pink-purple squares with no cellular structure in every class. GAN samples have speckled, noisy texture that looks broadly like tissue but lacks the glandular rings visible in real class-6 patches. Diffusion samples are the most varied and look right for adipose, background and lymphocyte rows. Its class-6 row, though, shows dark, blotchy purple patches that look more like debris or tumour than like regular mucosal glands.

In the feature-space scatter, CVAE samples sit in one small blob off to the side of the real data. The two GAN seeds each cover a different central region and miss the outer clusters. Diffusion samples spread over all three main real clusters, which matches its higher recall.

The label-agreement chart shows diffusion close to the real reference for most classes but far below it for classes 4 and 6. The class-balancing chart shows all four synthetic top-ups below the real-only seed-spread band, with resampling at the top edge of that band.

The CVAE loss curves flatten by about epoch 40. Diffusion validation loss is still falling

# Findings (cont.)

at epoch 40. The GAN discriminator loss sits near 2 throughout, while the generator loss stays low.

## How the models compare

All four models are scored in the same classifier feature space, so their fidelity metrics can be compared. The ranking by realism is cdiff_s0, then the two GAN seeds, then cvae_s0, and diffusion has much better coverage (recall 0.288 against at most 0.0583 for the others). For this objective, though, the metric that matters is whether the target-class images are recognisably class 6. On that measure, cgan_s0 did best (0.7179) and cdiff_s0 did poorly (0.2621). Even cgan_s0's images still lowered class-6 F1 in the downstream test (0.8971). Neither realism nor class-6 recognisability was enough to make synthetic images useful here.

## Answer to the objective

No. The success criterion was a class-6 F1 gain over real data plus resampling without lowering macro-F1. Every synthetic top-up fell short:

- real_plus_cvae_s0: 0.9028
- real_plus_cgan_s0: 0.8971
- real_plus_cgan_s1: 0.9
- real_plus_cdiff_s0: 0.9003
- real_plus_resampled (baseline): 0.9191

All four were lower by more than the 0.0063 noise level. Macro-F1 also dropped, from 0.9366 with resampling to between 0.9291 and 0.933. Resampling itself beat real-only training (0.9175) by less than the noise level. With a 1.634x imbalance and over 1,400 real class-6 training images, the classifier already handles class 6 about as well as the overall average.

# Downstream use of the generated data

## Top up class 6 (normal mucosa) with diffusion images

Verdict: NOT_RECOMMENDED | task: class_balancing | model: cdiff_s0

The diffusion model was the most realistic of the four overall, but adding its class-6 images lowered class-6 F1 below both resampling and the real-only runs by more than the noise level. The real-trained classifier recognised only about a quarter of its class-6 images as class 6, so most of the added images were mislabelled as mucosa.

- evidence: downstream.results.real_plus_cdiff_s0.target_f1 = 0.9003 (vs downstream.results.real_plus_resampled.target_f1 = 0.9191; downstream.noise_level_target_f1 = 0.0063)

- evidence: downstream.results.real_plus_cdiff_s0.macro_f1 = 0.9291 (vs downstream.results.real_plus_resampled.macro_f1 = 0.9366)

- evidence: cdiff_s0.conditional.per_class_agreement.6 = 0.2621 (vs cdiff_s0.conditional.per_class_heldout_accuracy.6 = 0.8974)

- caveat: Diffusion was still improving at its 40-epoch maximum (best epoch = last epoch), so a longer-trained model could do better; that was not tested here.

## Top up class 6 with conditional-GAN images

Verdict: NOT_RECOMMENDED | task: class_balancing | model: cgan_s0

Both GAN seeds lowered class-6 F1 relative to resampling, by more than the noise level. Whether the generated images were recognised as class 6 varied hugely between seeds, so the GAN is not a reliable source of class-6 images.

- evidence: downstream.results.real_plus_cgan_s0.target_f1 = 0.8971 (vs downstream.results.real_plus_resampled.target_f1 = 0.9191)

- evidence: downstream.results.real_plus_cgan_s1.target_f1 = 0.9 (vs downstream.results.real_plus_resampled.target_f1 = 0.9191)

- evidence: cgan_s0.conditional.per_class_agreement.6 = 0.7179 (vs cgan_s1.conditional.per_class_agreement.6 = 0.0798)

- caveat: cgan_s0 and cgan_s1 differ only in seed, yet one was mostly recognised as class 6 and the other almost never; this instability is itself the finding.

## Top up class 6 with conditional-VAE images

Verdict: NOT_RECOMMENDED | task: class_balancing | model: cvae_s0

The CVAE produced blurry, texture-free patches that the classifier never assigned to class 6. Adding them lowered class-6 F1 below resampling.

- evidence: downstream.results.real_plus_cvae_s0.target_f1 = 0.9028 (vs downstream.results.real_plus_resampled.target_f1 = 0.9191)

- evidence: cvae_s0.conditional.per_class_agreement.6 = 0 (vs cvae_s0.conditional.label_agreement_reference = 0.94)

# Downstream use of the generated data (cont.)

## Oversample real class-6 images (non-generative baseline)

Verdict: CONDITIONAL | task: class_balancing | model: real_plus_resampled

Resampling was the best option tested, but it beat real-only training by less than the noise level, so with this mild 1.634x imbalance it is about the same as doing nothing. Use it only if a balanced training set is wanted for other reasons; don't expect a measurable class-6 gain.

- evidence: downstream.results.real_plus_resampled.target_f1 = 0.9191 (vs downstream.results.real_seed0.target_f1 = 0.9175; downstream.noise_level_target_f1 = 0.0063)

- caveat: Measured on a 20,000-image stratified subsample with one small CNN trained for 20 epochs; the noise estimate comes from just two seeds.

## Not tested (no recommendation either way)

- Balancing class 4 (mucus), which is nearly as small as class 6; this would need target.class set to '4' and a rerun of the downstream test.

- Synthetic class-6 images from a diffusion model trained to convergence (the 20-minute budget stopped training at 40 epochs while validation loss was still falling); this would need a larger budget.

- Effect on a larger classifier or on the full 89,996-image training set, or on the external-centre CRC-VAL-HE-7K test split, which this project does not load.

- Use of the synthetic images for anything other than classifier training (e.g. sharing or teaching); no evaluation targeted those uses.

# Limitations

- Image realism is judged in the feature space of a classifier trained on the same real data; differences that do not affect classification (fine texture, staining details) are not measured by these numbers.

- The class-balancing test uses one classifier architecture and one seed per augmented training set; differences below the real-only seed spread (0.0063) are not meaningful.

- No independent critique was run for this report.

- Training used a 20,000-image stratified subsample of the 89,996 available images, chosen to fit the 20-minute budget. Results on the full data may differ.

- The diffusion model was undertrained: its validation loss was still falling at the 40-epoch cap. A longer run might produce better class-6 images; this was not tested.

- Whole-run wall-clock time was 23.58 minutes, over the 20-minute budget. Model training stayed inside the budget; the extra time was evaluation and the 4.02-minute downstream test, which the budget does not cover.

- The downstream classifier is one small CNN trained for 20 epochs. The noise level comes from only two seeds, so it is a rough yardstick, not a confidence interval.

- Class 4 (mucus) is almost as small as class 6. Choosing class 6 as "rarest" was the literal reading of the objective; balancing class 4 was not tested.

- In the CVAE latent-space figure, classes 6, 7 and 8 are drawn in the same grey and cannot be told apart. Its caption says labels were not used to train the model, which is inaccurate for a conditional VAE: its decoder is conditioned on the label. The figure is decorative here and the conclusions do not depend on it.

- The memorization checks measure distance to the nearest training image. They are evidence about copying, not a formal privacy guarantee.

# Reproducibility

Command: ./run.sh pathmnist "Balance the rarest tissue class to improve a classifier" --report-number 2

Plan seed 0; plan revision 1; budget 20 min; pipeline time 23.58 min

Preprocessing hash fa3206903097716f

python 3.13.5, torch 2.14.0, numpy 2.5.3, scikit-learn 1.9.1

- cvae_s0: seed 0, config hash 8dc688a92041778f, 1.794 min on mps

- cgan_s0: seed 0, config hash a5b8bc1ccff2123c, 3.087 min on mps

- cgan_s1: seed 1, config hash cf2bf05ab37136a1, 3.072 min on mps

- cdiff_s0: seed 0, config hash 4d5c1aa906c80647, 7.548 min on mps

Generated 2026-10-25 12:24. All intermediate files: outputs/pathmnist/.